Ageing and eating.

Epidemiological studies propose that extension of the human lifespan or the reduction of age associated diseases may be achieved by physical exercise, caloric restriction, and by consumption of certain substances such as resveratrol, selenium, flavonoids, zinc, omega 3 unsaturated fatty acids, vitamins E and C, Ginkgobiloba extracts, aspirin, green tea catechins, antioxidants in general, and even by light caffeine or alcohol consumption. Though intriguing, these studies only show correlative (not causative) effects between the application of the particular substance and longevity. On the other hand, obesity is yet a strong menace to the western society and it will emerge even more so throughout the next decades according to the prediction of the WHO. Although obesity is considered a severe problem, very little is known about the molecular mechanisms causing the associated degeneration of organs and finally death. Nutrient related adverse consequences for health and thus ageing may be due to a high sugar or high fat diet, excessive alcohol consumption and cigarette smoke amongst others. In this article we examine the interdependencies of eating and ageing and suggest yeast, one of the most successful ageing models, as an easy tool to elucidate the molecular pathways from eating to ageing. The conservation of most ageing pathways in yeast and their easy genetic tractability may provide a chance to discriminate between the correlative and causative effects of nutrition on ageing.